A complete proof of Theorem 3

The three-move super-expanders over $\mathbb{F}_2[t]/(t^\ell)$

September 8, 2026

Abstract

We complete the proof of Theorem 3 in the supplied draft *Some Banach Property (T) stuff*. The theorem holds with exactly its stated single-scale convexity hypothesis, and the Poincaré constant depends only on the stated deficit δ. The proof uses the draft's six-factor averaging construction, a characteristic-two strengthening of its Heisenberg contraction lemma, and a 14-factor elementary factorization over a finite local ring. In particular, no bounded-generation theorem over the polynomial ring, infinite-group property (T) theorem, or completeness assumption on the normed space is needed.

1 Statement and scope

Fix $\ell \geq 1$, and put

$$R_\ell = \mathbb{F}_2[t]/(t^\ell), \qquad I_\ell = tR_\ell, \qquad V_\ell = \{(x, y, z) \in R_\ell^3 : x \equiv 1 \pmod{t}\}.$$

Thus $|V_\ell| = 2^{3\ell-1}$. The three moves are

$$\begin{aligned} a(x, y, z) &= (x, y + x, z), \quad b(x, y, z) = (x, y, z + y), \\ c(x, y, z) &= (x + tz, y, z). \end{aligned} \tag{1.1}$$

They are involutions and preserve V_ℓ. Write $S = \{a, b, c\}$. Throughout, an edge is sampled by choosing $v \in V_\ell$ and $s \in S$ uniformly and using (v, sv). This is the three-move multigraph convention: loops and coinciding moves retain their prescribed multiplicities.

Theorem 1.1 (Theorem 3 of the draft, completed)**.** *The graph defined by* (1.1) *is connected for every $\ell \geq 1$. Let B be a real or complex normed vector space such that, for some $\delta > 0$,*

$$|u| = |v| = 1, \quad |u - v| \geq \tfrac{1}{2} \quad \Longrightarrow \quad \left|\frac{u + v}{2}\right| \leq 1 - \delta. \tag{1.2}$$

There is $c_\delta > 0$, depending only on δ, such that for every $\ell \geq 1$ and every $f : V_\ell \to B$,

$$\frac{1}{2}\mathbb{E}_{v \in V_\ell,\, s \in S}|f(sv) - f(v)|^2 \;\geq\; c_\delta \, \frac{1}{2}\mathbb{E}_{u,v \in V_\ell}|f(u) - f(v)|^2. \tag{1.3}$$

The choices of u and v on the right are independent and uniform.

Relation to the draft. The graph, norm hypothesis, and averaging construction are those of [1, Definition 1, Theorem 3, and equations (18)–(19)]. This note supplies the bridge from that construction to the three specific moves, proves the required convexity estimate at the correct scale, and checks the averaging calculation after omitting constant coefficients. The proof is self-contained; references to the draft identify its contributions rather than serve as unproved black boxes.

The case $B = \{0\}$ is immediate. Otherwise (1.2), applied to u and $-u$, implies $0 < \delta \leq 1$. We assume this henceforth.

2 The finite group behind the graph

Write $e_{ij}(p) = I + pE_{ij}$ for an elementary matrix. On column vectors,

$$a = e_{21}(1), \qquad b = e_{32}(1), \qquad c = e_{13}(t).$$

The relevant group is

$$Q_\ell = \{g \in \mathrm{SL}_3(R_\ell) : g \bmod t \text{ is lower unitriangular}\}. \tag{2.1}$$

It preserves V_ℓ. Call an elementary matrix *allowed* when

$$e_{ij}(p), \qquad \begin{cases} p \in R_\ell, & i > j, \\ p \in I_\ell, & i < j. \end{cases} \tag{2.2}$$

Lemma 2.1 (Generation with degree-uniform word bounds)**.** *Every allowed elementary matrix lies in* $\langle a, b, c\rangle$*. Moreover, for every fixed* D*, all matrices* $e_{ij}(p)$ *with* $p \in t\mathbb{F}_2[t]$ *and* $\deg p \le D$ *have word lengths in* a, b, c *bounded independently of* ℓ*.*

Proof. Use the convention $[g, h] = ghg^{-1}h^{-1}$. All the following identities hold over $\mathbb{F}_2[t]$, before taking any quotient:

$$\begin{aligned} [b, a] &= e_{31}(1), & [c, b] &= e_{12}(t), & [a, c] &= e_{23}(t), \\ [e_{23}(t), e_{31}(1)] &= e_{21}(t), & [e_{31}(1), e_{12}(t)] &= e_{32}(t), & [e_{32}(t), a] &= e_{31}(t). \end{aligned}$$

Together with c, these produce all six $e_{ij}(t)$. For distinct i, j, k, the elementary commutator relation gives

$$[e_{ik}(t^m), e_{kj}(t)] = e_{ij}(t^{m+1}).$$

Induction therefore produces all $e_{ij}(t^m)$, $m \ge 1$. Since $e_{ij}(p + q) = e_{ij}(p)e_{ij}(q)$, it produces every $e_{ij}(p)$ with $p \in t\mathbb{F}_2[t]$. The three lower constant matrices are $a, b, [b, a]$. Splitting off the constant coefficient proves generation.

For bounded degree, this construction uses a finite list of words over the polynomial ring. The same words work in every quotient R_ℓ, which gives the uniformity assertion. □

Lemma 2.2 (Fourteen elementary factors)**.** *Every element of* Q_ℓ *is a product of at most* 14 *allowed elementary matrices. Consequently* $Q_\ell = \langle a, b, c\rangle$*.*

Proof. Let $g \in Q_\ell$. Three lower elementary row operations eliminate the entries below the diagonal: first positions 21 and 31, then 32. Each pivot is 1 mod t and hence a unit. The resulting upper-triangular matrix has diagonal entries 1 mod t and upper entries in I_ℓ. Three upper elementary row operations, with coefficients in I_ℓ, clear those upper entries. Thus six allowed operations reduce g to a diagonal matrix.

For every unit $u \in R_\ell$, direct multiplication in characteristic two shows

$$\begin{pmatrix} u & 0 \\ 0 & u^{-1} \end{pmatrix} = e_{12}(u + 1)e_{21}(1)e_{12}(1 + u^{-1})e_{21}(u). \tag{2.3}$$

Here the elementary matrices are 2×2. Every unit is 1 mod t, so both upper coefficients lie in I_ℓ. For $d_1d_2d_3 = 1$, write

$$\mathrm{diag}(d_1, d_2, d_3) = \mathrm{diag}(d_1, d_1^{-1}, 1)\,\mathrm{diag}(1, d_1d_2, (d_1d_2)^{-1}).$$

Apply (2.3) in positions 12 and 23. This requires eight more allowed factors. Inverting the six row operations preserves the allowed form, so $3 + 3 + 8 = 14$ factors suffice. The generation assertion now follows from Lemma 2.1. □

Lemma 2.3 (Transitivity)**.** *The action $Q_\ell \curvearrowright V_\ell$ is transitive.*

Proof. For every $(x, y, z) \in V_\ell$, the matrix

$$\begin{pmatrix} x & 0 & 0 \\ y & x^{-1} & 0 \\ z & 0 & 1 \end{pmatrix} \in Q_\ell$$

takes $(1, 0, 0)$ to (x, y, z). □

Together, these lemmas prove connectivity, including the case $\ell = 1$. Notice that the bounded-generation argument is entirely over the finite local ring R_ℓ.

3 The correct convexity scale

For a finite probability space Ω, equip $L_2(\Omega; B)$ with

$$\|F\|_2^2 = \mathbb{E}_{\omega\in\Omega}|F(\omega)|^2.$$

The draft's Proposition 5 gives the following precise estimate. Its full proof is included in Appendix A.

Lemma 3.1 (Single-scale convexity of the L_2 sum)**.** *If B satisfies* (1.2)*, then $E = L_2(\Omega; B)$ satisfies*

$$\|\xi\| = \|\zeta\| = 1, \quad \|\xi - \zeta\| \geq 1 \quad \Longrightarrow \quad \left\|\frac{\xi + \zeta}{2}\right\| \leq 1 - \eta, \qquad \eta = \frac{\delta}{64}. \tag{3.1}$$

The separation threshold in (3.1) is 1, not $1/2$. The next lemma is why this is enough in characteristic two.

Lemma 3.2 (Characteristic-two Heisenberg contraction)**.** *Suppose a normed space E satisfies* (3.1)*. Let A, B, C be linear isometries such that*

$$A^2 = B^2 = C^2 = \mathrm{Id}, \qquad [A, B] = C, \qquad C \text{ commutes with } A, B.$$

Put $X = (\mathrm{Id} + A)/2$, $Y = (\mathrm{Id} + B)/2$, $Z = (\mathrm{Id} + C)/2$. Then

$$Z\zeta = 0 \quad \Longrightarrow \quad \|XY\zeta\| \leq (1 - \eta)\,\|\zeta\|\,. \tag{3.2}$$

Proof. Set $\omega = Y\zeta$. Then $B\omega = \omega$ and $C\omega = -\omega$. The relation $BA = CAB$ gives

$$BA\omega = CA\omega = -A\omega.$$

Thus the isometry B takes $\omega + A\omega$ to $\omega - A\omega$, so

$$\|\omega + A\omega\| = \|\omega - A\omega\|\,. \tag{3.3}$$

By the triangle inequality,

$$2\,\|\omega\| \leq \|\omega + A\omega\| + \|\omega - A\omega\| = 2\,\|\omega - A\omega\|\,.$$

If $\omega \neq 0$, (3.1) applies to $\omega/\|\omega\|$ and $A\omega/\|\omega\|$, yielding

$$\|XY\zeta\| = \left\|\frac{\omega + A\omega}{2}\right\| \leq (1 - \eta)\,\|\omega\| \leq (1 - \eta)\,\|\zeta\|\,.$$

The case $\omega = 0$ is immediate. □

This strengthens the draft's Lemma 7 in the case needed here. In general characteristic, annihilation by a central average supplies the weaker displacement used in the draft. Over $\mathbb{F}_2$, it supplies the exact relation $C\omega = -\omega$, which resolves the scale mismatch.

4 The averaging estimate, with positive degrees only

Fix a linear isometric representation π of Q_ℓ on a normed space E satisfying (3.1), and put $\rho = 1 - \eta$. A finitely supported probability measure on the group acts by its average under π, and hence acts as a contraction. We suppress π on such averages and use operator norm.

For $d \geq 1$, define

$$C_{ij}(d) = \operatorname{avg}_{\substack{p \in t\mathbb{F}_2[t] \\ \deg p \leq d}} \pi(e_{ij}(p \bmod t^\ell)). \tag{4.1}$$

The averaging set includes the zero polynomial. In contrast to the draft's averages, these omit the constant coefficient. All supporting matrices therefore belong to Q_ℓ, for all six root positions. When $d \geq \ell - 1$, this is the uniform average over $e_{ij}(I_\ell)$: every residue has the same number of polynomial representatives in the averaging set.

4.1 Pairing degrees in a Heisenberg subgroup

For a Heisenberg triple, write

$$x(p) = e_{12}(p), \qquad y(q) = e_{23}(q), \qquad z(r) = e_{13}(r),$$

and define

$$X_i = \frac{\mathrm{Id} + \pi(x(t^i))}{2}, \quad Y_j = \frac{\mathrm{Id} + \pi(y(t^j))}{2}, \quad Z_k = \frac{\mathrm{Id} + \pi(z(t^k))}{2}.$$

Set $X^{[1,r]} = X_r \cdots X_1$, and similarly for Y and Z; an empty product is the identity. Then $C_{12}(r) = X^{[1,r]}$, and similarly for the other two positions. All formulas are interpreted modulo t^ℓ. The argument works for every analogous triple of distinct matrix indices, with either order of its two noncentral roots: the possible sign in the commutator is immaterial over $\mathbb{F}_2$.

Lemma 4.1 (A one-degree change)**.** *For $r, s \geq 1$ and $N \geq 1$, put*

$$b_{r,s}(N) = \#\{(i, j) : 1 \leq i \leq r,\ 1 \leq j \leq s,\ i + j = N\}.$$

Then

$$\left\| X^{[1,r]} Z^{[1,N-1]} Y^{[1,s]} - X^{[1,r]} Z^{[1,N]} Y^{[1,s]} \right\| \leq 2\rho^{b_{r,s}(N)}. \tag{4.2}$$

Proof. Put $\Delta = Z^{[1,N-1]}(\mathrm{Id} - Z_N)$. The operator to be bounded is

$$X_r \cdots X_1 Y_s \cdots Y_1 \Delta.$$

The image of Δ is invariant under all the X_i, Y_j. On this image, every $z(t^k)$ with $1 \leq k < N$ acts trivially, and Z_N acts as zero. Consequently, whenever $i' + j < N$, the exact Heisenberg relation implies

$$X_{i'} Y_j \Delta = Y_j X_{i'} \Delta. \tag{4.3}$$

Indeed, one expands the two averages, and the central factor appearing when the two group elements are interchanged fixes $\operatorname{im} \Delta$.

Process the pairs (i, j) with $i + j = N$ in decreasing order of i. Move each Y_j leftward until it is immediately to the right of X_i. It passes only commuting Y factors and $X_{i'}$ with $i' < i$. The latter swaps are justified by (4.3), since $i' + j < N$. This makes all $b_{r,s}(N)$ matched factors $X_i Y_j$ adjacent.

Every factor preserves $\ker Z_N$. On that subspace, each matched pair contracts by ρ, by Lemma 3.2, because $[x(t^i), y(t^j)] = z(t^N)$. All unmatched factors are contractions. Finally $\|\Delta\| \leq 2$, since it is the difference of two contractions. This proves (4.2). □

In particular, this proof does not interpret a "commutator of averages" as an average of group commutators. The exact identity (4.3) is what the reordering uses.

4.2 The six-factor interpolation

Define, as in the draft but with (4.1),

$$g_n = C_{12}(n)C_{13}(2n)C_{23}(2n)C_{21}(2n)C_{31}(2n)C_{32}(n), \tag{4.4}$$
$$h_n = C_{23}(n)C_{13}(2n)C_{12}(2n)C_{32}(2n)C_{31}(2n)C_{21}(n), \tag{4.5}$$

for $n \geq 1$.

Proposition 4.2 (Successive averaging products are close)**.** *For every $n \geq 1$,*

$$\|g_n - h_{n+1}\|,\ \|h_n - g_{n+1}\| \leq (8n+12)\rho^{n-1} \leq 20n\rho^{n-1}. \tag{4.6}$$

Proof. For readability, abbreviate $C_{ij}(d)$ by $[ij]_d$. The complete interpolation from g_n to h_{n+1} is

$$\begin{aligned}
g_n &= [12]_n[13]_{2n}[23]_{2n}[21]_{2n}[31]_{2n}[32]_n \\
&\overset{1}{\approx} [12]_n[13]_{2n+2}[23]_{2n}[21]_{2n}[31]_{2n}[32]_n \\
&\overset{2}{\approx} [12]_n[13]_{2n+2}[23]_{2n}[21]_{2n}[31]_{2n+2}[32]_n \\
&\overset{3}{\approx} [12]_n[13]_{2n+2}[23]_{n+1}[21]_{2n}[31]_{2n+2}[32]_n \\
&\overset{4}{\approx} [12]_n[13]_{2n+2}[23]_{n+1}[21]_{n+1}[31]_{2n+2}[32]_n \\
&= [23]_{n+1}[13]_{2n+2}[12]_n[21]_{n+1}[31]_{2n+2}[32]_n \\
&= [23]_{n+1}[13]_{2n+2}[12]_n[32]_n[31]_{2n+2}[21]_{n+1} \\
&\overset{5}{\approx} [23]_{n+1}[13]_{2n+2}[12]_{2n+2}[32]_n[31]_{2n+2}[21]_{n+1} \\
&\overset{6}{\approx} [23]_{n+1}[13]_{2n+2}[12]_{2n+2}[32]_{2n+2}[31]_{2n+2}[21]_{n+1} = h_{n+1}.
\end{aligned} \tag{4.7}$$

Each approximate step changes only the middle factor of a Heisenberg triple. Apply Lemma 4.1 once for each degree crossed, and telescope. Factors outside the triple are contractions and do not increase the error. The complete accounting is as follows. The triple column lists (X, Z, Y), so its middle entry is the root being changed.

Step	Triple (X, Z, Y)	(r, s)	Degrees N crossed	$b_{r,s}(N) \geq$
1	$(12, 13, 23)$	$(n, 2n)$	$2n+1,\ 2n+2$	$n-1$
2	$(21, 31, 32)$	$(2n, n)$	$2n+1,\ 2n+2$	$n-1$
3	$(13, 23, 21)$	$(2n+2, 2n)$	$n+2, \ldots, 2n$	$n+1$
4	$(23, 21, 31)$	$(n+1, 2n+2)$	$n+2, \ldots, 2n$	$n+1$
5	$(13, 12, 32)$	$(2n+2, n)$	$n+1, \ldots, 2n+2$	n
6	$(12, 32, 31)$	$(2n+2, 2n+2)$	$n+1, \ldots, 2n+2$	n

For $n = 1$, steps 3 and 4 change nothing. The table follows directly from

$$b_{r,s}(N) = \max\{0,\ \min(r, N-1) - \max(1, N-s) + 1\}. \tag{4.8}$$

The two exact equalities in (4.7) exchange the noncentral factors of a Heisenberg triple. The resulting central coefficients have degree at most $n + (n + 1) = 2n + 1$ and zero constant term, so they are absorbed by the central average $C_{13}(2n + 2)$ or $C_{31}(2n + 2)$, respectively. Equivalently, translation of the uniformly averaged central parameter by their product preserves its distribution.

The six approximate steps cross respectively

$$2, \quad 2, \quad n-1, \quad n-1, \quad n+2, \quad n+2$$

degrees. Every one-degree error is at most $2\rho^{n-1}$. Their sum is $(8n + 12)\rho^{n-1}$.

For the other inequality, let J be the permutation matrix reversing the three coordinates. The automorphism

$$\theta(g) = J(g^{-1})^{\mathsf{T}}J$$

preserves Q_ℓ and sends $e_{ij}(p)$ to $e_{4-j,\,4-i}(p)$. It interchanges the group-algebra expressions defining g_n and h_n. Apply the inequality just proved to $\pi \circ \theta$ to obtain $\|h_n - g_{n+1}\| \le (8n + 12)\rho^{n-1}$. Finally, $8n + 12 \le 20n$ for $n \ge 1$. □

5 Finite stabilization and uniform almost-invariance

Let U_ℓ and L_ℓ denote the upper and lower unitriangular subgroups, respectively, whose off-diagonal entries lie in I_ℓ. Define the operator

$$P = \operatorname{avg}_{u\in U_\ell} \pi(u)\ \operatorname{avg}_{v\in L_\ell} \pi(v). \tag{5.1}$$

In particular,

$$\pi(u)P = P \qquad (u \in U_\ell). \tag{5.2}$$

No assertion that P is a projection is needed.

Set $f_n = g_n$ for odd n and $f_n = h_n$ for even n. Once $n \ge \max\{1, \ell - 1\}$, every elementary average in these products is uniform on its full subgroup $e_{ij}(I_\ell)$. Multiplication in either root order parametrizes U_ℓ bijectively by three such coefficients, and likewise for L_ℓ. For example,

$$e_{12}(p)e_{13}(r)e_{23}(q) = I + pE_{12} + qE_{23} + (r + pq)E_{13}, \qquad p, q, r \in I_\ell.$$

Since $pq \in I_\ell$, the change $r \mapsto r + pq$ is a bijection of I_ℓ. Hence, for all sufficiently large n,

$$g_n = h_n = f_n = P. \tag{5.3}$$

Thus Proposition 4.2 yields, by a *finite* telescoping sum,

$$\|f_N - P\| \le \tau_N, \qquad \tau_N := \sum_{n=N}^{\infty} 20n\rho^{n-1} = 20\rho^{N-1}\left(\frac{N}{\eta} + \frac{\rho}{\eta^2}\right). \tag{5.4}$$

The infinite series is only a numerical bound on that finite sum. In particular, $\tau_N \to 0$ at a rate depending only on δ.

We will also use all index relabelings of this construction. Conjugation by a permutation matrix need not preserve Q_ℓ, but it does preserve the principal congruence subgroup

$$K_\ell = \{g \in \mathrm{SL}_3(R_\ell) : g \equiv I \pmod{t}\} \subseteq Q_\ell,$$

which contains every factor used in (4.4)–(4.5). Therefore the same calculation applies to $\pi|_{K_\ell}$ after any permutation of indices. For every $i \ne j$, it produces operators $f_N^{(ij)}$ and $P^{(ij)}$ with

$$\left\|f_N^{(ij)} - P^{(ij)}\right\| \le \tau_N, \qquad \pi(e_{ij}(p))P^{(ij)} = P^{(ij)} \quad (p \in I_\ell). \tag{5.5}$$

Indeed, choose the relabeling so that ij is an upper root.

A uniform support-length bound. By Lemma 2.1, for each N there is a finite L_N, independent of ℓ and of the normed space, such that every supporting group element of every $f_N^{(ij)}$ has a word of length at most L_N in S. Only six elementary coefficients of degree at most $2N$ occur. An explicit choice, verified in Appendix C, is

$$L_N = 612 \cdot 4^N. \tag{5.6}$$

No optimization of this bound is required.

6 From almost-invariance to the Poincaré inequality

Take $E = L_2(V_\ell; B)$, with its normalized norm, and use the permutation representation

$$(\pi(g)F)(v) = F(g^{-1}v).$$

This is a linear isometric representation, and Lemma 3.1 supplies (3.1). Let $F \in E$ satisfy

$$\mathbb{E}_v F(v) = 0, \qquad \|F\|_E = 1,$$

and put

$$\varepsilon = \max_{s \in S} \|\pi(s)F - F\|_E .$$

A word of length m moves F by at most $m\varepsilon$, by telescoping and isometry. Averaging the words supporting $f_N^{(ij)}$ gives

$$\left\| f_N^{(ij)} F - F \right\|_E \le L_N \varepsilon.$$

Together with (5.5), this implies

$$\left\| P^{(ij)} F - F \right\|_E \le L_N \varepsilon + \tau_N.$$

For $u = e_{ij}(p)$ with $p \in I_\ell$, the vector $P^{(ij)}F$ is fixed by u. Consequently

$$\|\pi(u)F - F\|_E \le 2 \left\| P^{(ij)} F - F \right\|_E \le 2L_N \varepsilon + 2\tau_N. \tag{6.1}$$

For a lower elementary matrix with arbitrary coefficient, write $p = p_0 + p_+$, where $p_0 \in \mathbb{F}_2$ and $p_+ \in I_\ell$. The constant lower elementary matrices have word lengths at most 4: they are $a, b, [b, a]$. It follows that every allowed elementary matrix u satisfies

$$\|\pi(u)F - F\|_E \le (2L_N + 4)\varepsilon + 2\tau_N. \tag{6.2}$$

Using the 14-factor decomposition in Lemma 2.2, and telescoping once more,

$$\sup_{g \in Q_\ell} \|\pi(g)F - F\|_E \le (28L_N + 56)\varepsilon + 28\tau_N. \tag{6.3}$$

Transitivity gives

$$\operatorname{avg}_{g \in Q_\ell} \pi(g)F = \mathbb{E}_{v \in V_\ell} F(v) = 0,$$

where the right-hand side is viewed as a constant function. Average $F - \pi(g)F$ in (6.3) and use the triangle inequality to obtain

$$1 \le (28L_N + 56)\varepsilon + 28\tau_N.$$

Choose $N = N(\delta) \geq 1$ so that $\tau_N \leq 1/56$. Then

$$\max_{s\in S} \|\pi(s)F - F\|_E \geq \kappa_\delta, \qquad \kappa_\delta = \frac{1}{56L_{N(\delta)} + 112}. \tag{6.4}$$

This lower bound is uniform in ℓ and in B.

Now let $f : V_\ell \to B$ be arbitrary, put $\bar{f} = \mathbb{E}_v f(v)$, and set $F = f - \bar{f}$. If $\|F\|_E = 0$, the desired inequality is immediate. Otherwise apply (6.4) to $F/\|F\|_E$. Since a, b, c are involutions,

$$\begin{aligned}\mathbb{E}_{v,s}|f(sv) - f(v)|^2 &= \frac{1}{3}\sum_{s\in S} \|\pi(s)F - F\|_E^2 \\ &\geq \frac{\kappa_\delta^2}{3}\,\|F\|_E^2\,.\end{aligned}$$

On the other hand, the triangle inequality and $(r+s)^2 \leq 2r^2 + 2s^2$ give

$$\mathbb{E}_{u,v}|f(u) - f(v)|^2 \leq 4\mathbb{E}_v|f(v) - \bar{f}|^2 = 4\,\|F\|_E^2\,.$$

Combining these proves Theorem 1.1, with

$$\boxed{c_\delta = \frac{\kappa_\delta^2}{12} = \frac{1}{12\big(56L_{N(\delta)} + 112\big)^2}.} \tag{6.5}$$

The factors $1/2$ in (1.3) appear on both sides and cancel. All averages used to prove the inequality are finite, so completeness of B has never been used.

7 What must be added or adjusted in the draft

The graph-to-group bridge. Identify $\langle a, b, c\rangle$ with Q_ℓ, prove transitivity, and use the 14-factor decomposition from Section 2. This both resolves the connectivity question and supplies the bounded-generation statement needed for these specific graphs. The polynomial-ring bounded-generation input in the draft's infinite-group argument is unnecessary here.

The convexity threshold. Proposition 5 transfers the hypothesis on B to a separation-1 estimate on $L_2(V_\ell; B)$. It does not give the separation-$1/2$ estimate used in the stated general-field Lemma 7. The characteristic-two lemma in Section 3 is the required additional argument.

The averaging supports and commutator wording. Use positive-degree coefficients as in (4.1), so all six root averages lie in Q_ℓ. The degree counts change slightly, but Proposition 4.2 proves that the same bound $20n\rho^{n-1}$ remains valid. In the proof of the draft's Lemma 8, replace the informal assertion about commutators of averages by the exact identity (4.3).

Scope of the verification. These additions prove Theorem 3 without weakening its hypothesis and with no dependence of c_δ on the particular space B. They also justify the draft's footnote about other fixed thresholds below 2, as shown next. The separate infinite-group Theorem 6 is not needed for this verification.

A Proof of the L_2 convexity estimate

We prove Lemma 3.1, following the elementary argument in [1, Appendix A]. Let $F, G \in L_2(\Omega; B)$ have norm 1. Pointwise, write

$$F = ru, \qquad G = sv, \qquad r, s \geq 0, \qquad |u| = |v| = 1.$$

When a coefficient is zero, choose the corresponding unit vector arbitrarily. Define the nonnegative scalar functions

$$A = \frac{r+s}{2}, \qquad D = \frac{|r-s|}{2}, \qquad M = \min(r, s) = A - D,$$

and let $H = \mathbf{1}_{\{|u-v| \geq 1/2\}}$. Then

$$\|A\|_2^2 + \|D\|_2^2 = 1. \tag{A.1}$$

The triangle inequality gives the pointwise estimates

$$|F - G| \leq 2D + \tfrac{1}{2}M + 2HM, \tag{A.2}$$

$$\left|\frac{F+G}{2}\right| \leq A - \delta HM. \tag{A.3}$$

For the second inequality, for example when $r \geq s$, write the midpoint as $((r-s)/2)u + s(u+v)/2$ and apply (1.2) on the event $H = 1$.

Suppose now that $\|F - G\|_2 \geq 1$. Since $M \leq A$ and $\|A\|_2 \leq 1$, (A.2) implies

$$1 \leq 2\|D\|_2 + \tfrac{1}{2}\|M\|_2 + 2\|HM\|_2 \leq \tfrac{1}{2} + 2\|D\|_2 + 2\|HM\|_2 .$$

Thus

$$\|D\|_2 + \|HM\|_2 \geq \tfrac{1}{4}. \tag{A.4}$$

Also, because $A \geq M$ and $0 < \delta \leq 1$,

$$A^2 - (A - \delta HM)^2 = 2\delta HAM - \delta^2 HM^2 \geq \delta HM^2.$$

Squaring (A.3), averaging, and using (A.1) and (A.4), we obtain

$$\begin{aligned}\left\|\frac{F+G}{2}\right\|_2^2 &\leq 1 - \|D\|_2^2 - \delta\|HM\|_2^2 \\ &\leq 1 - \frac{\delta}{2}\big(\|D\|_2 + \|HM\|_2\big)^2 \\ &\leq 1 - \frac{\delta}{32}.\end{aligned}$$

Finally $\sqrt{1-x} \leq 1 - x/2$ for $0 \leq x \leq 1$ gives the claimed deficit $\eta = \delta/64$.

B Any fixed separation threshold below two

Suppose instead that, for fixed $0 < \alpha < 2$ and $\delta > 0$,

$$|u| = |v| = 1, \quad |u - v| \geq \alpha \quad \Longrightarrow \quad \left|\frac{u+v}{2}\right| \leq 1 - \delta. \tag{B.1}$$

Then the same graphs satisfy (1.3) with a positive constant depending only on α and δ. Here is the missing justification for the draft's footnote.

Put

$$\beta = \frac{\alpha + 2}{2} < 2, \qquad \eta_\alpha = \frac{\delta(2-\alpha)^2}{64}.$$

Repeat Appendix A, now with $H = \mathbf{1}_{\{|u-v| \geq \alpha\}}$. The pointwise difference estimate is

$$|F - G| \leq 2D + \alpha M + 2HM.$$

For unit F, G with $\|F - G\|_2 \geq \beta$, it yields

$$\|D\|_2 + \|HM\|_2 \geq \frac{\beta - \alpha}{2} = \frac{2 - \alpha}{4}.$$

The identical midpoint calculation gives

$$\|F\|_2 = \|G\|_2 = 1, \quad \|F - G\|_2 \geq \beta \quad \Longrightarrow \quad \left\| \frac{F+G}{2} \right\|_2 \leq 1 - \eta_\alpha. \tag{B.2}$$

In the Heisenberg configuration of Lemma 3.2, set

$$d = \|\omega - A\omega\| = \|\omega + A\omega\|,$$

using (3.3). The case $\omega = 0$ is immediate, so assume $\omega \neq 0$. If $d \geq \beta \|\omega\|$, apply (B.2) to obtain $\|X\omega\| \leq (1 - \eta_\alpha) \|\omega\|$. If $d < \beta \|\omega\|$, then directly

$$\|X\omega\| = \frac{d}{2} < \frac{\beta}{2} \|\omega\| \leq (1 - \eta_\alpha) \|\omega\|.$$

The last inequality holds because $\eta_\alpha \leq (2 - \alpha)/4 = 1 - \beta/2$. Thus Lemma 3.2 remains true with η_α in place of η. All subsequent arguments apply verbatim with $\rho = 1 - \eta_\alpha$. This proves the extension.

C An explicit support-length bound

The constructions in Lemma 2.1 give the six $e_{ij}(t)$ with word lengths at most $1, 4, 4, 16, 16, 34$. Upper bounds W_m on the lengths of all six $e_{ij}(t^m)$ are therefore given by

$$W_1 = 34, \qquad W_{m+1} = 2W_m + 68, \qquad W_m = 51 \cdot 2^m - 68.$$

For $p \in t\mathbb{F}_2[t]$ of degree at most D, factor $e_{ij}(p)$ into its nonzero monomials. Its word length is at most

$$\sum_{m=1}^{D} W_m = 102 \cdot 2^D - 102 - 68D \leq 102 \cdot 2^D.$$

Every supporting word of f_N or any index relabeling uses six such matrices of degree at most $2N$. Thus $L_N = 612 \cdot 4^N$ is valid. Together with (5.4) and (6.5), this makes the Poincaré constant completely explicit.

References

[1] *Some Banach Property (T) stuff*. Incomplete manuscript dated June 28, 2026, supplied as `banach-property-t(7).pdf`, 8 pages.